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SEntFiN 1.0: Entity-Aware Sentiment Analysis for Financial News

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Fine-grained financial sentiment analysis on news headlines is a challenging task requiring human-annotated datasets to achieve high performance. Limited studies have tried to address the sentiment extraction task in a setting where multiple entities are present in a news headline. In an effort to further research in this area, we make publicly available SEntFiN 1.0, a human-annotated dataset of 10,753 news headlines with entity-sentiment annotations, of which 2,847 headlines contain multiple entities, often with conflicting sentiments. We augment our dataset with a database of over 1,000 financial entities and their various representations in news media amounting to over 5,000 phrases. We propose a framework that enables the extraction of entity-relevant sentiments using a feature-based approach rather than an expression-based approach. For sentiment extraction, we utilize 12 different learning schemes utilizing lexicon-based and pre-trained sentence representations and five classification approaches. Our experiments indicate that lexicon-based n-gram ensembles are above par with pre-trained word embedding schemes such as GloVe. Overall, RoBERTa and finBERT (domain-specific BERT) achieve the highest average accuracy of 94.29% and F1-score of 93.27%. Further, using over 210,000 entity-sentiment predictions, we validate the economic effect of sentiments on aggregate market movements over a long duration.

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32 Pages

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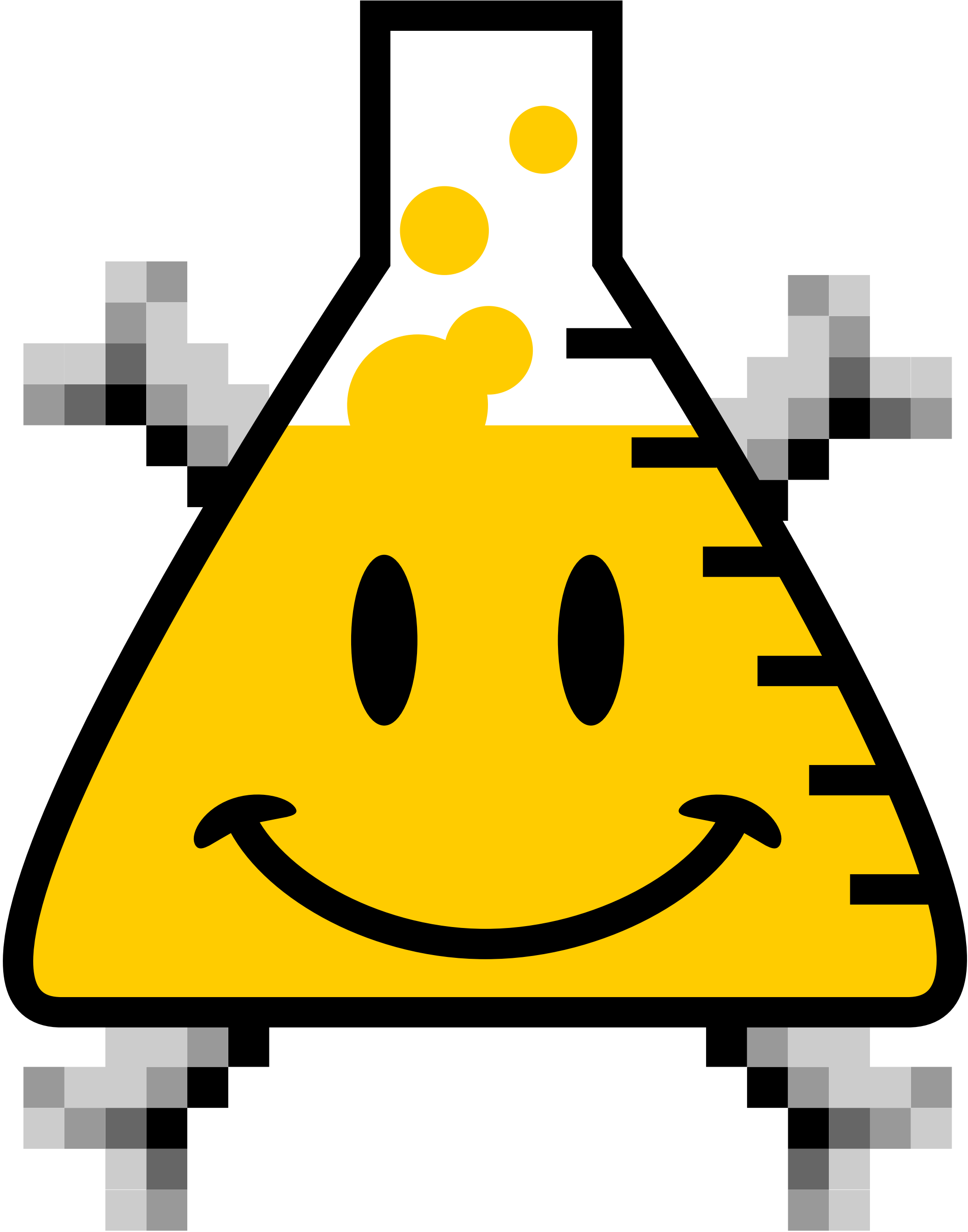
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